Dynamic causal effects II

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Plan for this slide deck

1. Quantifying the importance of shocks
2. Examples of shock measurement in the applied literature
3. Dealing with measurement error in the shock
4. Small-sample bias of LP
5. Panel LP

How important are climate shocks?

- Consider again the Bilal & Känzig (2026) application from the previous slide deck.
- We have estimated the dynamic causal effects (i.e., IRF) of world GDP wrt. climate shocks.
- But world GDP depends on many other kinds of shocks, too.
- We would like to quantify how *important* our (residualized) shock $\tilde{x}_t$ is for the outcome y_t, relative to everything else that drives y_t.

Forecast error variance decomposition

- **Forecast error variance decomposition** (FEVD): how much would the h-step-ahead MSFE $E[(y_{t+h} - y_{t+h|t})^2]$ decrease if we hypothetically knew the intervening shocks $\tilde{x}_t, \ldots, \tilde{x}_{t+h}$ in advance?
- Estimation procedure (separately for each horizon h):
  1. Regress y_{t+h} on all control variables $\boldsymbol{r}_t, \boldsymbol{w}_{t-1}, \ldots, \boldsymbol{w}_{t-p}$. Compute the (forecast) residual $e_{t+h|t}$.
  2. $FEVD_h \equiv R^2$ from regression of $e_{t+h|t}$ on $\tilde{x}_t, \tilde{x}_{t+1}, \ldots, \tilde{x}_{t+h}$.
- Note: this definition depends on the set of controls. We're asking how well current+future $\tilde{x}_t$ explain the forecast error from the regression with baseline controls.

```
# Leads of residualized shock
for l in range(max_horizon + 1):
    df[f'temp_anom_resid_lead{l}'] = df['temp_anom_resid'].shift(-l)

# Estimate FEVD
fevd = np.empty(max_horizon + 1)
for h in range(max_horizon + 1):
    y = df['l_gdp'].shift(-h)
    reg1 = sm.OLS(y, sm.add_constant(df[control_cols]), missing='drop').fit()
    cols = [f'temp_anom_resid_lead{l}' for l in range(h + 1)]
    reg2 = sm.OLS(reg1.resid,
                  sm.add_constant(df.loc[reg1.resid.index, cols]),
                  missing='drop').fit()
    fevd[h] = reg2.rsquared
```

h	FEVD
0	0.017150
1	0.023892
2	0.061400
3	0.113741
4	0.160317
5	0.226175
6	0.302844
7	0.324311
8	0.273270
9	0.267634
10	0.266006

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0	0.017150
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Climate shocks explain 32% of GDP forecast variance 7 years out!

Examples of shock measurement in the applied literature

1. Romer & Romer (1989, 2004) combine information from the meeting transcripts of the Federal Open Market Committee (FOMC) with data on the Fed's expectations of future economic conditions to isolate the "exogenous" components of changes in the federal funds rate.
2. Gertler & Karadi (2015) measure a monetary shock as the high-frequency movement in futures prices in a 30-minute time window around FOMC announcements.
3. Miranda-Agrippino, Hacıoğlu-Hoke & Bluwstein (2025) measure productivity news shocks using patent application data.

- All these are likely to contain some **measurement error**: errors in interpreting the Fed meetings, financial micro-structure noise, patents as imperfect measures of ideas, etc.

Proxies

- Suppose we have constructed a direct measure z_t of an underlying shock ε_{1t}, but it's contaminated with measurement error ξ_t:

$$z_t = \gamma \varepsilon_{1t} + \xi_t.$$

  We call z_t a **proxy** or **external instrument** for ε_{1t}.

- Proxy assumptions:

  1. **Relevance:** $\gamma \neq 0$. The proxy is informative about ε_{1t}.
  2. **Dynamic exogeneity:** $\text{Cov}(\xi_t, \varepsilon_{js}) = 0$ for all j, s. The noise ξ_t is unrelated to any systematic economic shocks.

Proxies: attenuation bias

$$z_t = \gamma \varepsilon_{1t} + \xi_t$$

- Recall the representation we derived in the previous slide deck:

$$y_{t+h} = \mu + \Theta_{h1} \varepsilon_{1t} + v_{h,t+h}.$$

- A simple LP of y_{t+h} on z_t suffers from **attenuation bias**:

$$\hat{\beta}_h \equiv \frac{\widehat{\text{Cov}}(y_{t+h}, z_t)}{\widehat{\text{Var}}(z_t)} \xrightarrow{p} \frac{\text{Cov}(y_{t+h}, z_t)}{\text{Var}(z_t)} = \Theta_{h1} \underbrace{\frac{\gamma \, \text{Var}(\varepsilon_{1t})}{\gamma^2 \, \text{Var}(\varepsilon_{1t}) + \text{Var}(\xi_t)}}_{\text{bias factor}},$$

  since $\text{Cov}(v_{h,t+h}, \xi_t) = \text{Cov}(\varepsilon_{1t}, \xi_t) = 0$ for classical meas. error.

- Intuitively, a 1-unit increase in z_t does not correspond to a 1-unit increase in the unobserved shock ε_{1t} – there is also noise.

- Problem: the bias factor is not estimable.

Proxies: attenuation bias (cont.)

- Solution to the bias problem: normalize the scale of the shock using another observed variable $\bar{y}_t$ (the **normalization** variable).
- If $\bar{y}_t$ also satisfies an SVMA model, then

$$\bar{y}_t = \bar{\mu} + \bar{\Theta}_{01}\varepsilon_{1t} + \bar{v}_{0,t},$$

  where $\bar{\Theta}_{01}$ is the impact impulse response of $\{\bar{y}_t\}$ wrt. $\{\varepsilon_{1t}\}$.
- Let $\hat{\beta}_{h,z\to y}$ be the coefficient in an LP of y_{t+h} on z_t (previous slide).
- Let $\hat{\beta}_{0,z\to \bar{y}}$ be the coefficient in an LP of $\bar{y}_t$ on z_t.
- Then, by the calculation on the previous slide,

$$\frac{\hat{\beta}_{h,z\to y}}{\hat{\beta}_{0,z\to \bar{y}}} \xrightarrow{p} \frac{\Theta_{h1} \times (\text{bias factor})}{\bar{\Theta}_{01} \times (\text{bias factor})} = \frac{\Theta_{h1}}{\bar{\Theta}_{01}},$$

  since the bias factor depends only on the proxy, not the outcome.

Proxies: unit effect normalization

$$\frac{\hat{\beta}_{h,z\to y}}{\hat{\beta}_{0,z\to \bar{y}}} \xrightarrow{p} \frac{\Theta_{h1}}{\bar{\Theta}_{01}}$$

- Conclusion: the ratio $\hat{\beta}_{h,z\to y}/\hat{\beta}_{0,z\to \bar{y}}$ of two LP coefficients estimates the *relative* impulse response $\Theta_{h1}/\bar{\Theta}_{01}$.
- $1/\bar{\Theta}_{01}$ = how big of a shock ε_{1t} you need in order to raise $\bar{y}_t$ by 1 unit on impact.
- $\Theta_{h1}/\bar{\Theta}_{01}$ = causal effect on y_{t+h} of a shock ε_{1t} of a magnitude that raises $\bar{y}_t$ by 1 unit (on impact).
  - This normalization of the magnitude of the shock is called the **unit effect normalization**.
- Unit-effect-normalized impulse responses are convenient to interpret.

LP-IV

$$\frac{\hat{\beta}_{h,z\to y}}{\hat{\beta}_{0,z\to \bar{y}}} \xrightarrow{p} \frac{\Theta_{h1}}{\bar{\Theta}_{01}}$$

- We have seen that we can estimate unit-effect-normalized impulse responses as the ratio of two LPs: $z \to y$ and $z \to \bar{y}$.
- Rather than explicitly running two LPs, notice that

$$\frac{\hat{\beta}_{h,z\to y}}{\hat{\beta}_{0,z\to \bar{y}}} = \frac{\widehat{\text{Cov}}(y_{t+h}, z_t)/\widehat{\text{Var}}(z_t)}{\widehat{\text{Cov}}(\bar{y}_t, z_t)/\widehat{\text{Var}}(z_t)} = \frac{\widehat{\text{Cov}}(y_{t+h}, z_t)}{\widehat{\text{Cov}}(\bar{y}_t, z_t)}.$$

  This is precisely the 2SLS estimator in a regression of the outcome y_{t+h} on the normalization variable $\bar{y}_t$, using the proxy z_t as an IV!

- The 2SLS estimator of the unit-effect-normalized impulse response is called **LP-IV**.

LP-IV: control variables and SE

- We can allow the proxy to be correlated with lagged control variables:

$$z_t = \gamma\varepsilon_{1t} + \boldsymbol{\delta}_1'\boldsymbol{w}_{t-1} + \cdots + \boldsymbol{\delta}_p'\boldsymbol{w}_{t-p} + \xi_t.$$

- Then we just need to control for $\boldsymbol{w}_{t-1}, \ldots, \boldsymbol{w}_{t-p}$ in our 2SLS regressions.
- If we control sufficiently for lags, we can use the conventional heteroskedasticity-robust standard errors for 2SLS.

LP-IV: Bilal & Känzig application

- It's reasonable to expect that the temperature anomalies are measured with error.
- We can estimate unit-effect-normalized impulse responses by LP-IV.
- Normalization variable $\bar{y}_t \equiv$ global temperature. So the scale of the climate shock is such that it raises global temperature by 1°C on impact.

First-stage regression ($z_t \to \bar{y}_t$):

OLS Regression Results

Dep. Variable:	temp	R-squared:	0.962
Model:	OLS	Adj. R-squared:	0.953
Method:	Least Squares	F-statistic:	177.6
Date:	Sat, 25 Apr 2026	Prob (F-statistic):	8.09e-34
Time:	15:47:44	Log-Likelihood:	78.687
No. Observations:	58	AIC:	-133.4
Df Residuals:	46	BIC:	-108.6
Df Model:	11		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
const	13.9362	0.104	133.909	0.000	13.732	14.140
temp_anom	0.6336	0.076	8.316	0.000	0.484	0.783
l_gdp_lag1	-0.0179	0.009	-2.079	0.038	-0.035	-0.001
l_gdp_lag2	0.0234	0.008	2.903	0.004	0.008	0.039
temp_anom_lag1	-0.2369	0.089	-2.653	0.008	-0.412	-0.062
temp_anom_lag2	0.3137	0.115	2.733	0.006	0.089	0.539
lnpoil_wti_lag1	0.0006	0.000	1.303	0.193	-0.000	0.002
lnpoil_wti_lag2	0.0003	0.000	0.618	0.537	-0.001	0.001
treasury1y_lag1	-0.0088	0.008	-1.062	0.288	-0.025	0.007
treasury1y_lag2	-0.0093	0.009	-1.068	0.285	-0.026	0.008
recessiondates_lag1	-0.0326	0.023	-1.396	0.163	-0.078	0.013
recessiondates_lag2	-0.0952	0.028	-3.442	0.001	-0.149	-0.041

First-stage regression ($z_t \to \bar{y}_t$):

OLS Regression Results

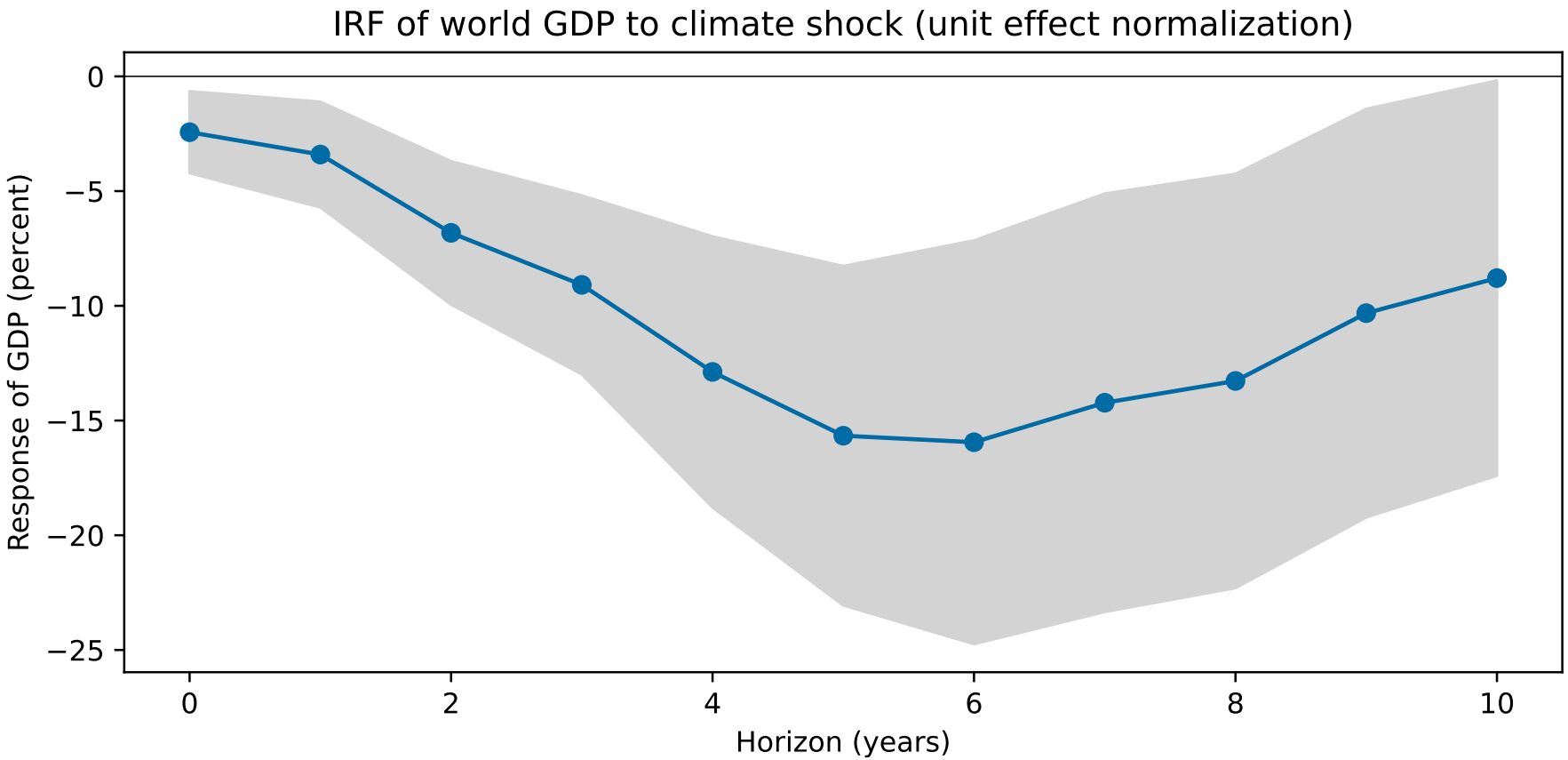
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	coef	st				
const	13.9362					
temp_anom	0.6336					
l_gdp_lag1	-0.0179					
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One-unit temperature anomaly causes 0.63°C rise in global temp

$\Longrightarrow$ scale outcome LPs by 1/0.63, or just run 2SLS

```
def local_projection_iv(df, outcome, norm_var, proxy, controls, max_horz):
    """LP-IV: instrument norm_var with proxy."""
    irf = np.empty(max_horz + 1)
    se  = np.empty(max_horz + 1)
    for h in range(max_horz + 1):
        y = df[outcome].shift(-h)
        exog = sm.add_constant(df[controls])
        reg = IV2SLS(dependent=y, exog=exog,
                     endog=df[[norm_var]], instruments=df[[proxy]]
                     ).fit(cov_type='robust')
        irf[h] = reg.params[norm_var]
        se[h]  = reg.std_errors[norm_var]
    return irf, se
```


IRF of world GDP to climate shock (unit effect normalization)
Response of GDP (percent)
0
−5
−10
−15
−20
−25
0
2
4
6
8
10
Horizon (years)

Narrative restrictions

- How to construct proxies in practice? One way is through qualitative **narrative restrictions**.
- Suppose we have information about a subset of historical dates where there were especially large *positive* or *negative* shocks of a certain kind (e.g., monetary policy shocks). At other dates we're unsure.
- Define (noisy) proxy:

$$z_t \equiv \begin{cases} +1 & \text{if large, positive shock at time } t, \\ -1 & \text{if large, negative shock at time } t, \\ 0 & \text{otherwise.} \end{cases}$$

  - Relevance: z_t is positively correlated with the shock of interest.
  - Dynamic exogeneity: z_t is uncorrelated with the other economic shocks if (i) these other shocks do not influence which dates we have information on, and (ii) misclassifications are random.

Weak instruments

- If the normalization variable $\bar{y}_t$ is only weakly correlated with the proxy z_t (conditional on controls), then we have a weak instrument.
- There are two reasons why the IV could be weak:
  1. The proxy z_t is dominated by noise, so only weakly correlated with the shock of interest ε_{1t}.
  2. The normalization variable $\bar{y}_t$ responds only weakly to the shock of interest ε_{1t} on impact.
- We should pick a normalization variable that correlates strongly with the proxy.

Small-sample bias of LP

- Though the LP estimator is consistent under our assumptions, it can still be biased in small samples, similar to VAR estimators.
- There exist various procedures for bias-correcting LP.
- Bootstrap procedures can also improve the finite-sample coverage probability of the LP confidence interval.
- For details, see the review article by Montiel Olea et al. (2026) on the reading list.

Dynamic causal effects in panel data

- Sometimes we have access to time series for many different economic **units**, like countries, sectors, firms, households, etc.
- Let y_{it} and x_{it} be an outcome and a shock for unit i at time t, where $i = 1, \ldots, n$ and $t = 1, \ldots, T$.
- A **panel LP** estimates the impulse response of y wrt. x at horizon h using a panel fixed effects regression

$$y_{i,t+h} = \hat{\nu}_i + \hat{\tau}_t + \hat{\beta}_h x_{it} + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{i,t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{i,t-p} + \hat{v}_{h,i,t+h}.$$

  - $\hat{\nu}_i$: unit fixed effects. These control for unobserved time-invariant heterogeneity across units.
  - $\hat{\tau}_t$: time fixed effects. These control for unobserved global factors that affect all units equally.
  - $\boldsymbol{w}_{it}$: vector of unit-specific control variables.

Dynamic causal effects in panel data (cont.)

- The regression on the previous slide assumes that the IRF is the same for all units.
- Suppose instead we want to model the impulse responses as depending on an observed unit characteristic s_i.
- We can model cross-unit heterogeneity in impulse responses using an interaction term:

$$y_{i,t+h} = \hat{\nu}_i + \hat{\tau}_t + \hat{\beta}_h x_{it} + \hat{\varphi}_h (s_i - \bar{s}) x_{it} + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{i,t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{i,t-p} + \hat{v}_{h,i,t+h},$$

where $\bar{s} \equiv \frac{1}{n} \sum_{i=1}^{n} s_i$.

- Then the impulse response estimate for unit i is

$$\tilde{\beta}_h(s_i) \equiv \hat{\beta}_h + \hat{\varphi}_h(s_i - \bar{s}).$$

Panel data: hypothetical example

$$y_{i,t+h} = \hat{\nu}_i + \hat{\tau}_t + \hat{\beta}_h x_{it} + \hat{\varphi}_h (s_i - \bar{s}) x_{it} + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{i,t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{i,t-p} + \hat{v}_{h,i,t+h}$$

- y_{it}: inflation rate of country i at time t
- x_{it}: monetary shock in country i at time t
- $\boldsymbol{w}_{it}$: real GDP, inflation, and monetary shock in country i at time t
- s_i: index of how independent the central bank in country i is from the country's government
- $\hat{\beta}_h$: estimated average impulse response (across countries)
- $\hat{\varphi}_h$ measures whether monetary shocks have larger effect on inflation in countries with more independent central banks

Panel data: aggregate shock

- In some panel data applications, the shock x_{it} is common to all units i, so $x_{it} = x_t$.
- Example: panel data regressions in Bilal & Känzig (2026).
  - y_{it}: log real GDP in country i at time t.
  - x_t: *global* temperature anomaly.
- Since x_t does not vary across units, it is perfectly multi-collinear with the time dummies. We therefore have to drop the latter if we want to identify the causal effects of x_t.
- Panel regression with aggregate shock:

$$y_{i,t+h} = \hat{\nu}_i + \hat{\beta}_h x_t + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{i,t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{i,t-p} + \hat{v}_{h,i,t+h}.$$

Panel data: aggregate shock (cont.)

$$y_{i,t+h} = \hat{\nu}_i + \hat{\beta}_h x_t + \hat{\boldsymbol{\delta}}_1' \boldsymbol{w}_{i,t-1} + \cdots + \hat{\boldsymbol{\delta}}_p' \boldsymbol{w}_{i,t-p} + \hat{v}_{h,i,t+h}$$

- $\boldsymbol{w}_{it}$ should include both unit-specific controls (e.g., y_{it}) and aggregate controls (e.g., x_t).
  - Include the same aggregate controls you would use if it were a pure time series LP.
- We could include an interaction term $(s_i - \bar{s}) \times x_t$ to estimate heterogeneous responses.
- Standard errors should be clustered by time to allow for additional (unobserved) aggregate shocks. Need to control for sufficient lags p, as in pure time series LP.

```
# Load country panel data
df_panel = pd.read_csv('data/bilal_kaenzig2026_panel.csv')
df_panel = df_panel[df_panel['year'] >= 1960].copy()
df_panel = df_panel.set_index(['country_code', 'year']).sort_index()

# Avg temperature by country
df_panel['avgtemp'] = df_panel.groupby('country_code')['lctmp_bkly_pw'].transform('mean')

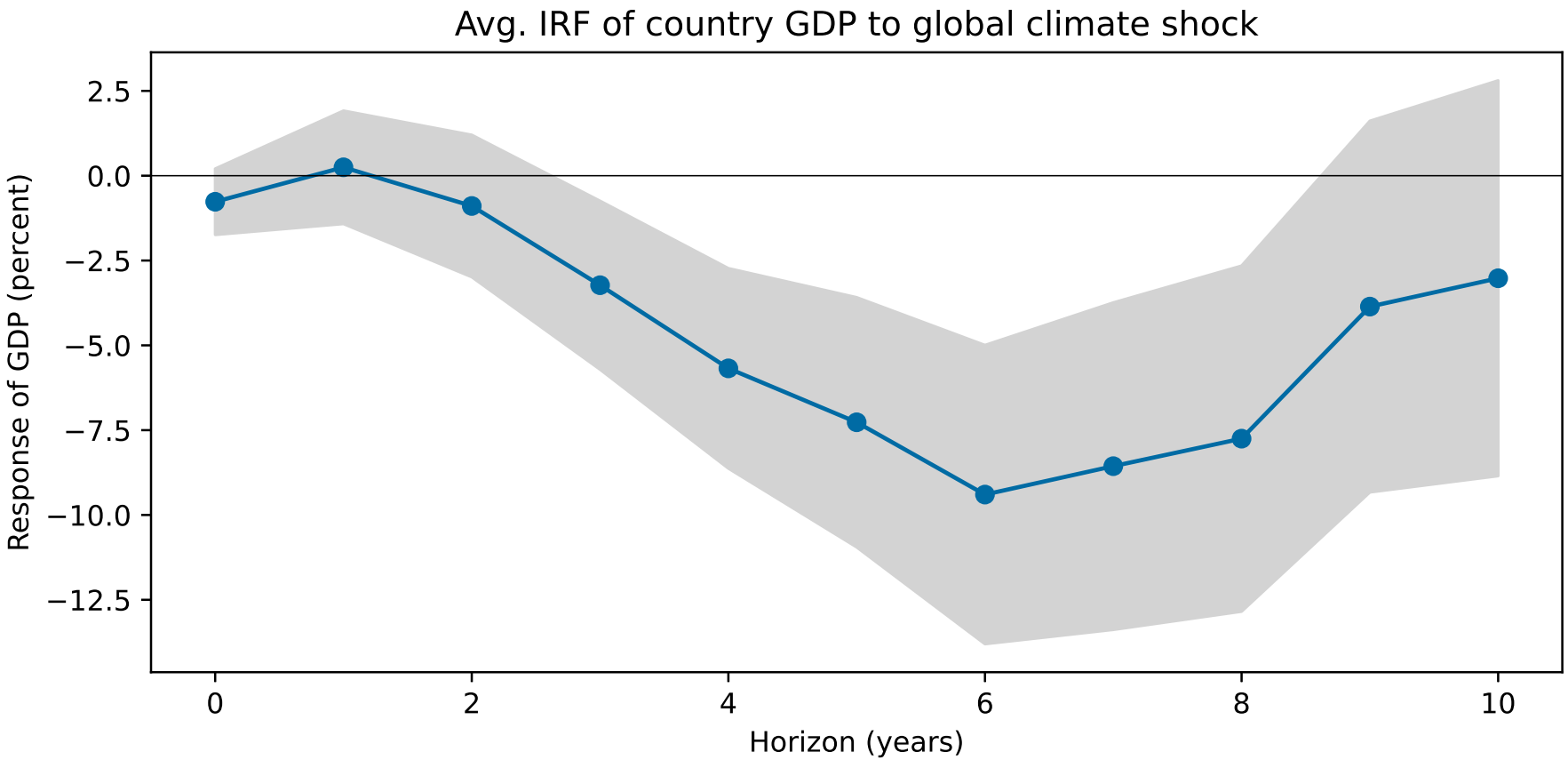
# Own lags of outcome (within country)
for l in range(1, nlag + 1):
    df_panel[f'lnrgdppc_pwt_lag{l}'] = (
        df_panel.groupby(level='country_code')['lnrgdppc_pwt'].shift(l)
    )

# Merge in aggregate shock + controls
df_panel = df_panel.join(df[['temp_anom'] + control_cols], on='year')
```

```
# Interaction term
avgtemp_global = df_panel.groupby(level='country_code')['avgtemp'].first().mean()
df_panel['avgtemp_d'] = df_panel['avgtemp'] - avgtemp_global # de-mean
df_panel['temp_anom_x_avgtemp'] = df_panel['temp_anom'] * df_panel['avgtemp_d']

# Panel LP with unit FE and time-clustered SE
own_lag_cols = [f'lnrgdppc_pwt_lag{l}' for l in range(1, nlag + 1)]
panel_regs = own_lag_cols + ['temp_anom', 'temp_anom_x_avgtemp'] + control_cols

irf_panel = np.empty(max_horizon + 1)
se_panel  = np.empty(max_horizon + 1)
inter_panel = np.empty(max_horizon + 1)
inter_se_panel = np.empty(max_horizon + 1)
for h in range(max_horizon + 1):
    y = df_panel.groupby(level='country_code')['lnrgdppc_pwt'].shift(-h)
    reg = PanelOLS(y, df_panel[panel_regs], entity_effects=True
                   ).fit(cov_type='clustered', cluster_time=True)
    irf_panel[h] = reg.params['temp_anom']
    se_panel[h]  = reg.std_errors['temp_anom']
    inter_panel[h] = reg.params['temp_anom_x_avgtemp']
    inter_se_panel[h] = reg.std_errors['temp_anom_x_avgtemp']
```


Avg. IRF of country GDP to global climate shock
Response of GDP (percent)
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
0
2
4
6
8
10
Horizon (years)

h	beta	beta_se	phi	phi_se
0	-0.768	0.979	-0.243	0.160
1	0.247	1.677	-0.406	0.257
2	-0.892	2.106	-0.538	0.327
3	-3.228	2.498	-0.570	0.349
4	-5.678	2.965	-0.624	0.389
5	-7.268	3.702	-0.720	0.417
6	-9.399	4.429	-0.801	0.412
7	-8.561	4.848	-0.722	0.423
8	-7.750	5.124	-0.749	0.411
9	-3.858	5.503	-0.822	0.359
10	-3.023	5.861	-0.799	0.348

`beta`: avg. impulse response
`phi`: coef. on interaction $(s_i - \bar{s}) \times x_t$

What did we learn?

- FEVD
- Identification from proxy variables
  - Measurement error
  - LP-IV
  - Narrative proxies
- LP small-sample bias
- Panel LP
  - Idiosyncratic vs. aggregate shocks
  - Interaction terms

Looking ahead

- If we believe the data is well described by a VAR model, can we use that to improve our estimates of IRFs and FEVDs?
- How do LP and VAR estimates compare?
- Yet more ways of identifying unobserved shocks